# EPIK8s at BTF

From Process Variables to Operational Interfaces

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# Why this matters?

- BTF systems expose a large amount of real-time data through EPICS PVs
- Operators and beam line scientists need to quickly understand system state and react accordingly
- Raw data alone is not sufficient for safe and efficient operations
- Interfaces must support fast decision-making under time pressure

# The problem

- EPICS exposes many PVs, often without structure from a user perspective
- Interfaces can become cluttered with too many values and no clear hierarchy
- Different systems may use inconsistent conventions and layouts
- Users are forced to interpret raw values instead of meaningful states

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| Instance Name | Status | PV Count | Connected | Event Rate | Data Rate (GB/day) | Engine write thread(s) | Max ETL(%) |
|---|---|---|---|---|---|---|---|
| appliance0 | Working | 1459 | 563 | 131.2 | 0.2 | 0.02 | 0.02 |

# Mental model

MONITORING

## Real Hardware

(power supply, diagnostic system, motors...)

- Produces physical signals and responds to control actions
- Not directly accessible to users or applications
- Requires a control layer to interface with software

## IOCs

- Software layer interfacing with the hardware
- Read data and sends commands to the device
- Run inside a container managed by EPIK8s
- Translate hw-level signals into PVs
- Ensure safe and controlled interaction with the system

## PVs

- EPICS abstraction presenting data and control points
- Expose system information (RB, states, alarms, ...)
- Allow control action (setpoints, commands)
- Contact between Control System and application

## GUI

- Application used by operator and users
- Reads PV to display system state and values
- Write to PV to send commands and setpoints
- Trasform raw data in meaning full info
- Ensure clarity and usability

CONTROL

# Recap on Pv types

- **Readback (RBV):** represents measured or actual values (ai, bi, waveform records)
- **Setpoint (SET):** represents desired or commanded values (ao, bo, stringout)
- **State (STATE):** indicates system status - e.g. ON/OFF (bi, mbbi)
- **Mode (MODE):** defines operational mode - e.g. LOCAL/REMOTE (mbbo, mbbi)
- **Alarm (ALARM):** indicates severity or abnormal conditions (**NOT separate PVs by default)**

# A real BTF example

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| PV Name | Status | Appliance | Connected? | Monitored? | Sampling period | Last event | Details | Quick chart |
|---|---|---|---|---|---|---|---|---|
| BTF2:daq:DAQ_BTF2:gun_status | Being archived | appliance0 | true | true | 1.0 | Mar/24/2026 09:34:55 UTC | | |
| BTF2:daq:DAQ_BTF2:mod_status | Being archived | appliance0 | true | true | 1.0 | Mar/24/2026 09:34:55 UTC | | |
| BTF2:daq:DAQ_BTF2:particle_status | Being archived | appliance0 | true | true | 1.0 | Mar/24/2026 06:37:41 UTC | | |
| BTF2:slow:CaloV:value | Being archived | appliance0 | true | true | 1.0 | Mar/26/2026 17:09:16 UTC | | |
| BTF2:slow:EH1_CaloV:value | Being archived | appliance0 | true | true | 1.0 | Mar/18/2026 07:37:01 UTC | | |
| BTF2:slow:EH1_calo_autotrack:value | Being archived | appliance0 | true | true | 1.0 | Mar/18/2026 07:36:49 UTC | | |
| BTF2:slow:EH2:ele_instant:value | Being archived | appliance0 | true | true | 1.0 | Mar/18/2026 07:36:59 UTC | | |
| BTF2:slow:EH2_CaloV:value | Being archived | appliance0 | true | true | 1.0 | Mar/24/2026 09:30:17 UTC | | |
| BTF2:slow:EH2_N_ele_tot:value | Being archived | appliance0 | true | true | 1.0 | Mar/18/2026 07:37:01 UTC | | |
| BTF2:slow:EH2_calo_autotrack:value | Being archived | appliance0 | true | true | 1.0 | Mar/18/2026 07:37:07 UTC | | |
| BTF2:slow:FITPIX:XCenter | Being archived | appliance0 | true | true | 1.0 | Mar/24/2026 09:56:37 UTC | | |
| BTF2:slow:FITPIX:XSTD | Being archived | appliance0 | true | true | 1.0 | Mar/24/2026 09:56:37 UTC | | |
| BTF2:slow:FITPIX:YCenter | Being archived | appliance0 | true | true | 1.0 | Mar/24/2026 09:56:37 UTC | | |
| BTF2:slow:FITPIX:YSTD | Being archived | appliance0 | true | true | 1.0 | Mar/24/2026 09:56:37 UTC | | |
| BTF2:slow:FITPIX:mean_hit | Being archived | appliance0 | true | true | 1.0 | Mar/24/2026 09:56:37 UTC | | |
| BTF2:slow:FITPIX:mean_sum | Being archived | appliance0 | true | true | 1.0 | Mar/24/2026 09:56:37 UTC | | |
| BTF2:slow:GaussProbe:DHRTB102:Error_source | Being archived | appliance0 | true | true | 1.0 | Mar/18/2026 07:37:06 UTC | | |
| BTF2:slow:GaussProbe:DHRTB102:GPBIP | Being archived | appliance0 | true | true | 1.0 | Mar/18/2026 07:37:13 UTC | | |
| BTF2:slow:GaussProbe:DHRTB102:value | Being archived | appliance0 | true | true | 1.0 | Never | | |
| BTF2:slow:GaussProbe:DHSTB001:Error_source | Being archived | appliance0 | true | true | 1.0 | Mar/18/2026 07:37:03 UTC | | |
| BTF2:slow:GaussProbe:DHSTB001:GPBIP | Being archived | appliance0 | true | true | 1.0 | Mar/18/2026 07:37:10 UTC | | |
| BTF2:slow:GaussProbe:DHSTB001:value | Being archived | appliance0 | true | true | 1.0 | Mar/26/2026 17:10:54 UTC | | |
| BTF2:slow:GaussProbe:DHSTB002:Error_source | Being archived | appliance0 | true | true | 1.0 | Mar/18/2026 07:36:51 UTC | | |
| BTF2:slow:GaussProbe:DHSTB002:GPBIP | Being archived | appliance0 | true | true | 1.0 | Mar/18/2026 07:37:09 UTC | | |
| BTF2:slow:GaussProbe:DHSTB002:value | Being archived | appliance0 | true | true | 1.0 | Mar/18/2026 07:36:50 UTC | | |

*EPICS Archiver Appliance Version 2.1.2*

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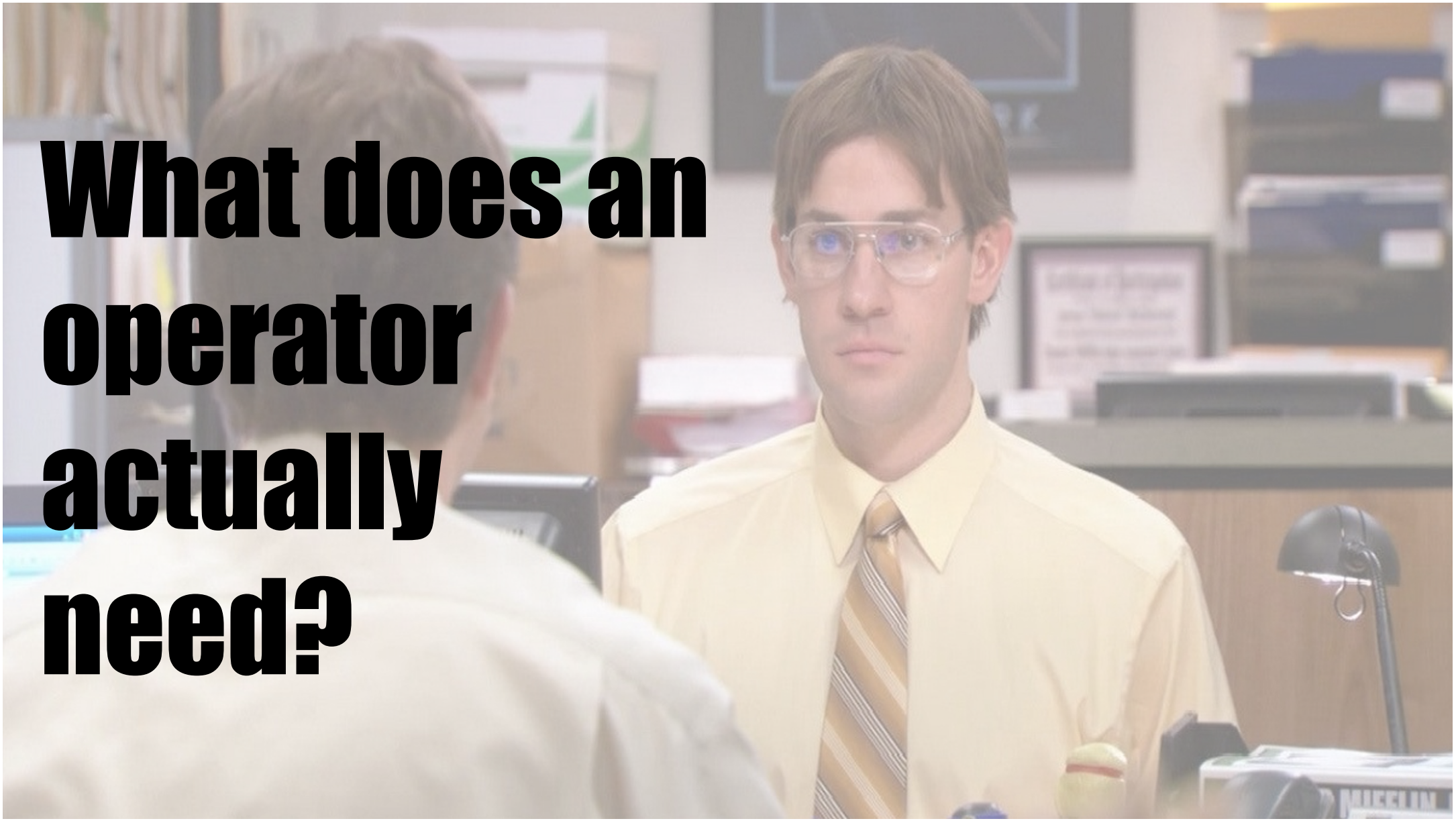
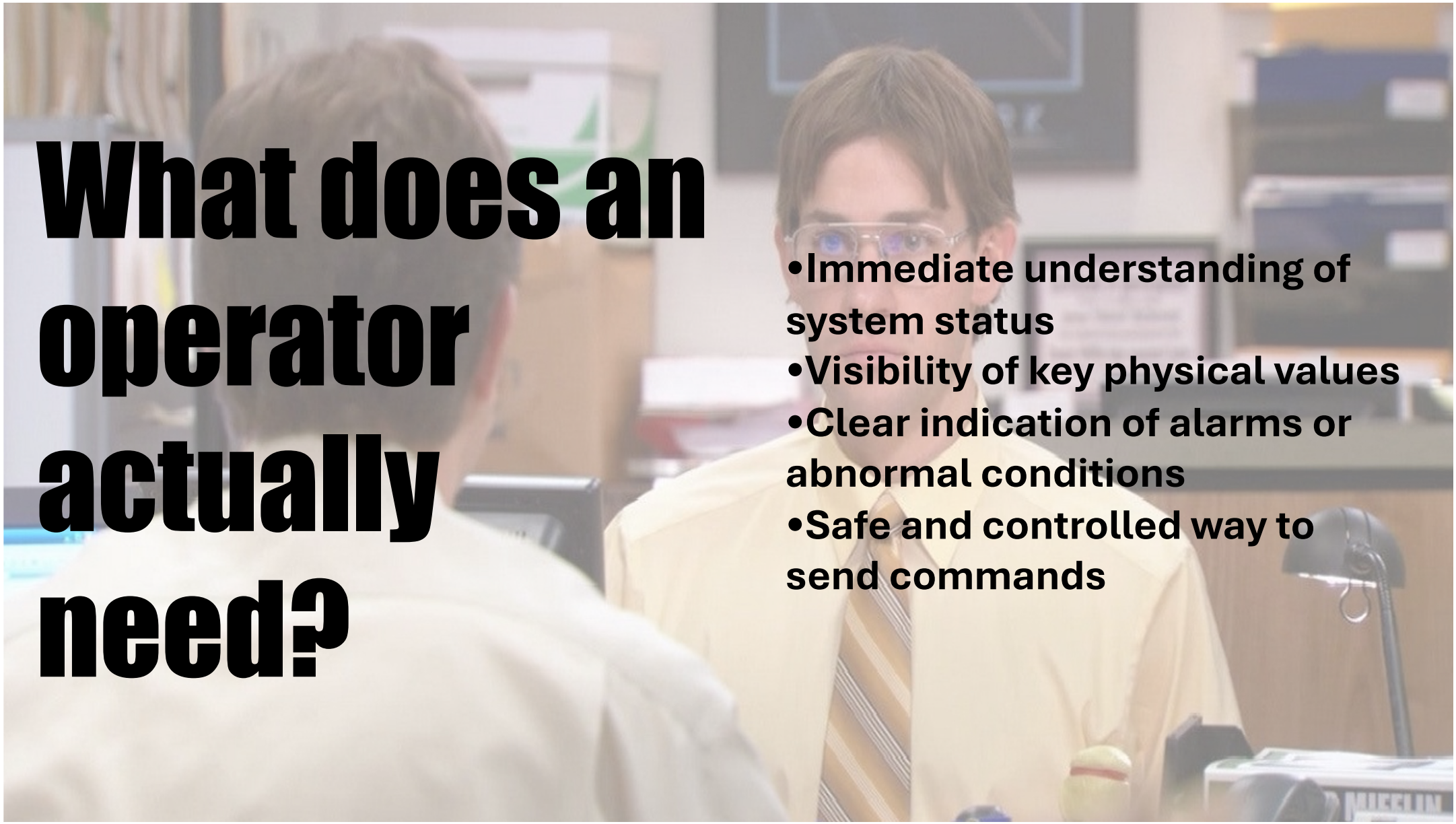
| PV Name | Status | Appliance | Connected? | Monitored? | Sampling period | Last event | Details | Quick chart |
|---|---|---|---|---|---|---|---|---|
| BTF2:slow:VUGPADME:value | Being archived | appliance0 | true | true | 1.0 | Mar/26/2026 17:10:53 UTC | | |
| BTF2:slow:VUGTB003:error_code | Being archived | appliance0 | true | true | 1.0 | Mar/23/2026 08:54:26 UTC | | |
| BTF2:slow:VUGTB003:error_status | Being archived | appliance0 | true | true | 1.0 | Never | | |
| BTF2:slow:VUGTB003:status | Being archived | appliance0 | true | true | 1.0 | Mar/23/2026 08:54:26 UTC | | |
| BTF2:slow:VUGTB003:timestamp | Being archived | appliance0 | true | true | 1.0 | Mar/26/2026 17:10:53 UTC | | |
| BTF2:slow:VUGTB003:value | Being archived | appliance0 | true | true | 1.0 | Mar/23/2026 08:54:26 UTC | | |
| BTF2:slow:VUGTB004:error_code | Being archived | appliance0 | true | true | 1.0 | Mar/18/2026 07:37:11 UTC | | |
| BTF2:slow:VUGTB004:error_status | Being archived | appliance0 | true | true | 1.0 | Never | | |
| BTF2:slow:VUGTB004:status | Being archived | appliance0 | true | true | 1.0 | Mar/23/2026 05:50:17 UTC | | |
| BTF2:slow:VUGTB004:timestamp | Being archived | appliance0 | true | true | 1.0 | Mar/26/2026 17:10:53 UTC | | |
| BTF2:slow:VUGTB004:value | Being archived | appliance0 | true | true | 1.0 | Mar/23/2026 05:50:17 UTC | | |
| BTF2:slow:VUGTB201:error_code | Being archived | appliance0 | true | true | 1.0 | Mar/18/2026 07:37:13 UTC | | |
| BTF2:slow:VUGTB201:error_status | Being archived | appliance0 | true | true | 1.0 | Never | | |
| BTF2:slow:VUGTB201:status | Being archived | appliance0 | true | true | 1.0 | Mar/23/2026 08:39:06 UTC | | |
| BTF2:slow:VUGTB201:timestamp | Being archived | appliance0 | true | true | 1.0 | Mar/26/2026 17:10:53 UTC | | |
| BTF2:slow:VUGTB201:value | Being archived | appliance0 | true | true | 1.0 | Mar/23/2026 08:39:06 UTC | | |
| BTF2:slow:VUGTB202:error_code | Being archived | appliance0 | true | true | 1.0 | Mar/23/2026 09:07:12 UTC | | |
| BTF2:slow:VUGTB202:error_status | Being archived | appliance0 | true | true | 1.0 | Never | | |
| BTF2:slow:VUGTB202:status | Being archived | appliance0 | true | true | 1.0 | Mar/23/2026 09:07:12 UTC | | |
| BTF2:slow:VUGTB202:timestamp | Being archived | appliance0 | true | true | 1.0 | Mar/26/2026 17:10:53 UTC | | |
| BTF2:slow:VUGTB202:value | Being archived | appliance0 | true | true | 1.0 | Mar/23/2026 09:07:12 UTC | | |
| BTF:MAG:FAST:PS:OutputCurrent | Being archived | appliance0 | true | true | 1.0 | Mar/26/2026 09:44:27 UTC | | |
| BTF:MAG:FAST:PS:OutputVoltage | Being archived | appliance0 | true | true | 1.0 | Mar/26/2026 09:44:27 UTC | | |
| BTF:MAG:FAST:PS:STATE_RB | Being archived | appliance0 | true | true | 1.0 | Mar/26/2026 09:44:29 UTC | | |
| BTF:TML:SLTTB001D:ACCL_RB | Being archived | appliance0 | false | true | 1.0 | Never | | |

*EPICS Archiver Appliance Version 2.1.2*


What does an
operator
actually
need?



# What does an operator actually need?

- Immediate understanding of system status
- Visibility of key physical values
- Clear indication of alarms or abnormal conditions
- Safe and controlled way to send commands

# Control Interface Design Principles

## Fast Interpretation
Understandable in < 1 sec

## Focus on Decisions
Show what matters

## Reusable patterns
Same PV → same behavior

## Standardized Structure
Predictable layout

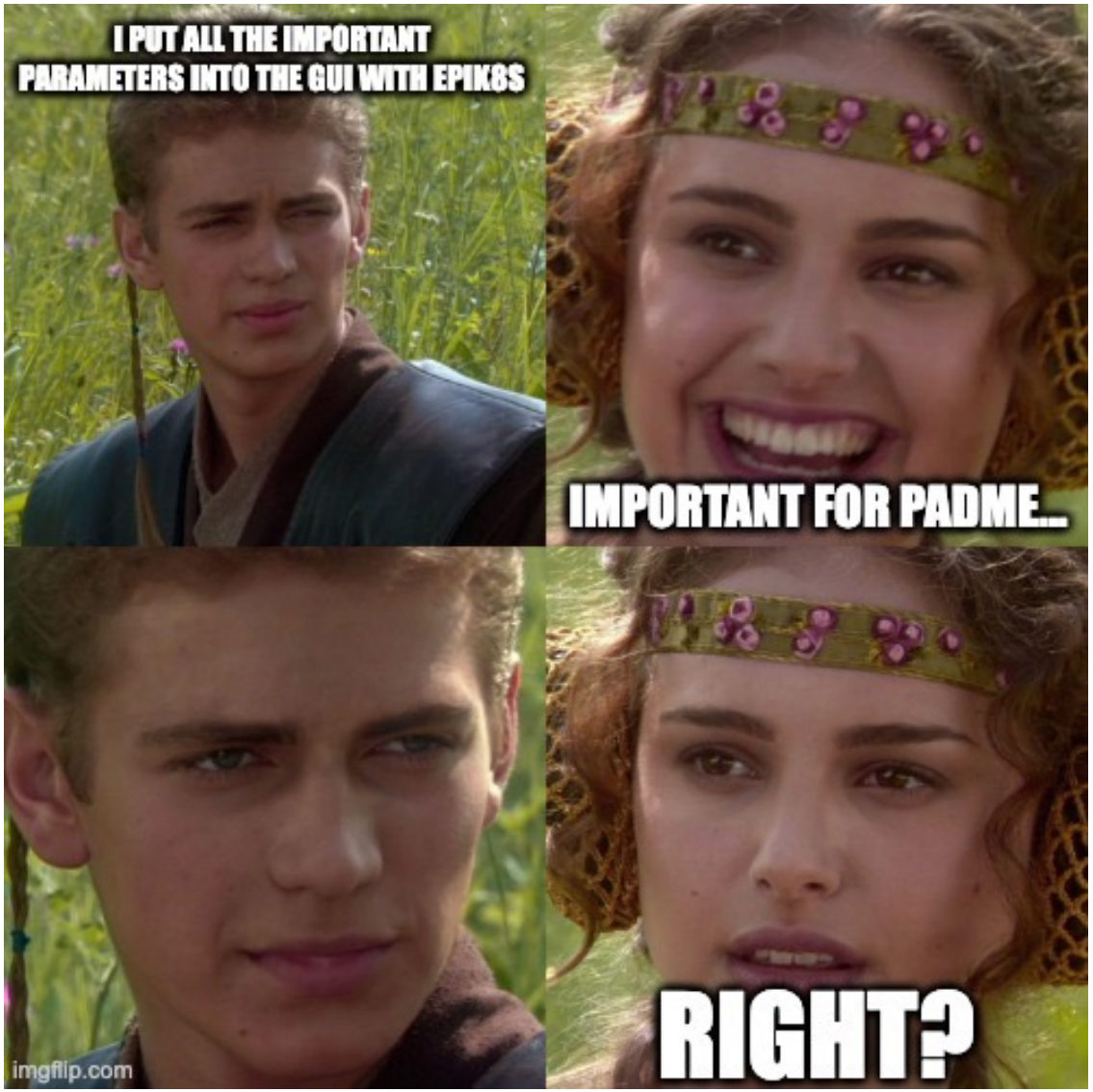
# Some ideas...

- Monitor the Humidity, Pressure and Temperature of the EH1 and EH2
  - BTF2:slow:PTUBTFEH1_Temp:value, BTF2:slow:PTUBTFEH1_Press:value, BTF2:slow:PTUBTFEH1_RH:value
  - BTF2:slow:PTUBTFEH2_Temp:value, BTF2:slow:PTUBTFEH2_Press:value, BTF2:slow:PTUBTFEH2_RH:value
- Monitor the values of the Hall probes installed on the magnet
  - BTF2:slow:GaussProbe:DHSTB001:value, BTF2:slow:GaussProbe:DHSTB201:value, BTF2:slow:GaussProbe:DHSTB202:value
- Monitor the vacuum level
  - BTF2:slow:VUGTB003:value, BTF2:slow:VUGTB004:value, BTF2:slow:VUGTB201:value, BTF2:slow:VUGTB202:value, BTF2:slow:VUGPADME:value
- Monitor the current of the the magnets
  - DAFNE:BTF:QUATB101:current_rb, DAFNE:BTF:QUATB102:current_rb, (...)
  - DAFNE:BTF:DHSTB001:current_rb, DAFNE:BTF:DHPTB101:current_rb, (...)
  - DAFNE:BTF:CHHTB001:current_rb, DAFNE:BTF:CVVTB001:current_rb, (...)

# The final task...

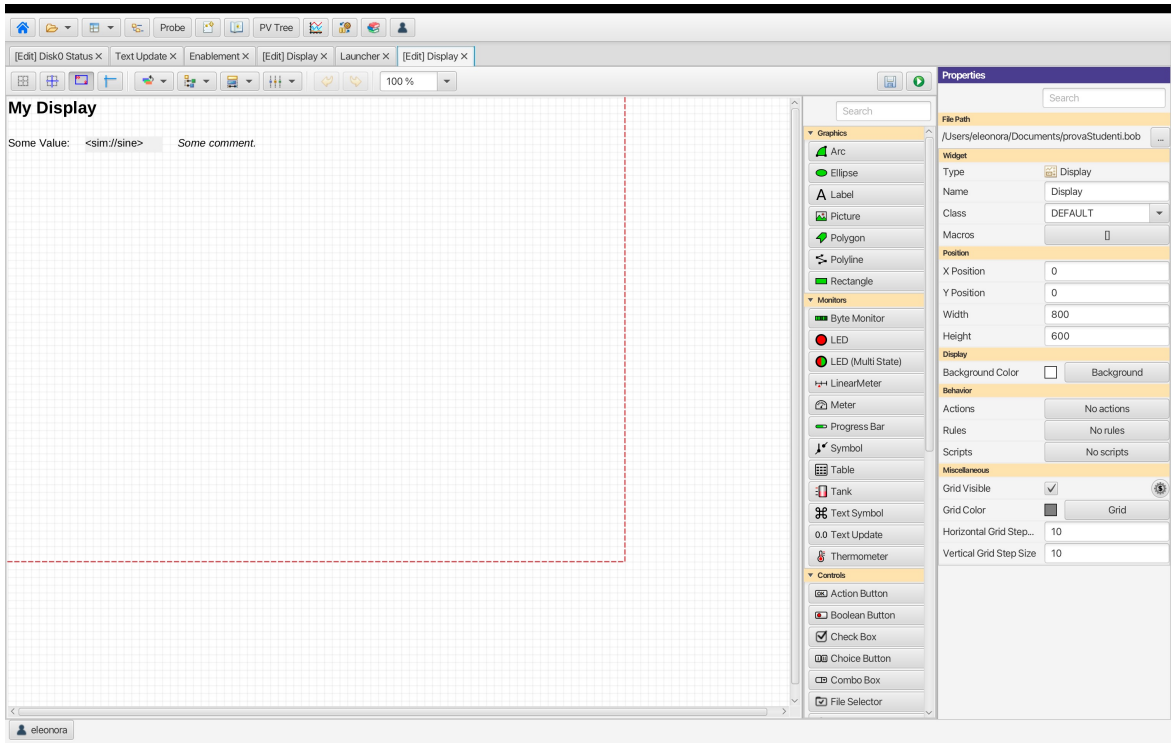
Let’s realize together a GUI to show all the pivotal PV needed by PADME

**THINK ABOUT WHAT USER WANT TO SEE**



# The exercise…

1. Check if the PV is archived on the [EPICS archiver]
2. Check the graph to understand the range of the PV
3. Open a new Display on Phoebus




- Use text update to see the value of a given PV changing (eventually) over the time
- Use LED, Stripcharts, … to give a graphical representation of the PV
- Open the terminal phoebus -s settings.ini –resource DoctoLab.bob